as complementary base pairing. The DNA double helix or duplex is held together by two forces,

a) Hydrogen bonding between complementary base pairs

b) Base-stacking interactions

The complementary between the DNA strands is attributable to the hydrogen bonding between base pairs but the base stacking interactions are largely non-specific, make the major contribution to the stability of the double helix.

14.5.3 Types of RNA molecules

Ribonucleic acids are similar to DNA. Cells contain up to eight times high quantity of RNA than DNA. RNA is found in large amount in the cytoplasm and a lesser amount in the nucleus. In the cytoplasm it is mainly found in ribosomes and in the nucleus, it is found in nucleolus.

RNA molecules are classified according to their structure and function into three major types

i. Ribosomal RNA (rRNA) ii. Messenger RNA (mRNA)

iii. Transfer RNA (tRNA)

rRNA

rRNA is mainly found in cytoplasm and in ribosomes, which contain 60% RNA and 40% protein. Ribosomes are the sites at which protein synthesis takes place.

tRNA

tRNA molecules have lowest molecular weight of all nucleic acids. They consist of 73 – 94 nucleotides in a single chain. The function of tRNA is to carry amino acids to the sites of protein synthesis on ribosomes.

mRNA

mRNA is present in small quantity and very short lived. They are single stranded, and their synthesis takes place on DNA. The synthesis of mRNA from DNA strand is called transcription. mRNA carries genetic information from DNA to the ribosomes for protein synthesis. This process is known as translation

Table 14.3 Difference between DNA and RNA

DNA	RNA
It is mainly present in nucleus, mitochondria and chloroplast	It is mainly present in cytoplasm, nucleolus and ribosomes
It contains deoxyribose sugar	It contains ribose sugar
Base pair A = T. G ≡ C	Base pair A = U. C ≡ G
Double stranded molecules	Single stranded molecules
It's life time is high	It is Short lived
It is stable and not hydrolysed easily by alkalis	It is unstable and hydrolyzed easily by alkalis
It can replicate itself	It cannot replicate itself. It is formed from DNA.

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